

Supporting Statement

Research Assistant – ART-transition Study

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Motivation and Suitability

The ART-transition study brings together participant support, longitudinal research and rigorous data practice in a way that closely matches both my experience and the direction in which I want to develop. Supporting participants consistently over 24 months, maintaining reliable study records and turning complex remote-monitoring data into defensible research findings are not separate responsibilities: each depends on the quality of the others. My background in health data science, mental-health-related research, database management and multidisciplinary study delivery would allow me to contribute strongly across this connected workflow.

Particularly compelling is the study's focus on the transition from adolescence into adulthood for people with ADHD. This is a period in which sustained engagement, clear communication and sensitivity to individual circumstances are essential. Although my present role is primarily data-intensive, it is grounded in participant-derived health information and in the principle that research quality begins with respectful treatment of the people whose experiences the data represent. The opportunity to combine that foundation with direct participant support, debrief interviews and meaningful PPIE is therefore a natural and purposeful next step.

Research Background and Mental Health Relevance

My academic background combines a BSc in Computer Engineering, assessed as equivalent to a UK 2:1, with an MSc with Merit in Health Data Science from the University of Exeter. Across four years of health and biomedical research, this training has been applied to study design, clinical and cohort data, statistical and machine-learning analysis, research databases, reproducible methods and scientific communication.

Current work at Dementias Platform UK (DPUK) involves supporting research across genomic, multi-omic, longitudinal and clinical-context data from multiple dementia cohorts. Alongside these responsibilities, I conceived a precision dementia subtyping project and developed the research idea and analytical protocol with guidance and collaboration from Dr Esteban Gomez and Professor Alfredo Iacoangeli at King's College London. The project examines whether integrated biological, clinical and longitudinal information can identify meaningful molecular or progression subtypes. Its methods include cross-cohort harmonisation, assessment of missingness and batch effects, feature engineering, clustering, pathway analysis and validation against clinical outcomes. A first-author poster was presented at the Alzheimer's Research UK Conference in 2026, and a protocol manuscript is in preparation.

This work sits within a broader interest in neurological and psychiatric heterogeneity. My polygenic-risk workflow supports research across dementia and related neurological and psychiatric conditions, while an earlier clinical machine-learning study used ethically approved hospital data to investigate mortality among patients with COVID-19 and diabetes. That study involved data cleaning, feature engineering, model evaluation, interpretation and manuscript development and led to a peer-reviewed publication. During my MSc placement with GSK Human Genetics, I also managed an independent UK Biobank whole-genome sequencing project within a regulated, international research environment. Together, these experiences demonstrate sustained scientific research, responsible work with sensitive health data and the ability to carry analyses through to publication and presentation.

Longitudinal Data and Database Management

The data requirements of ART-transition are especially well matched to my current responsibilities. DPUK receives heterogeneous data from different cohorts, technologies and documentation systems, so preparing it for research requires more than routine processing. File and table structures must be inspected, identifiers reconciled, missingness and metadata completeness assessed, inconsistencies investigated, provenance retained and limitations communicated clearly. Validation checks are designed to expose problems before analysis, while assumptions and decisions are recorded so that downstream researchers can understand how the data should be interpreted.

Database work is similarly embedded in my role. Using PostgreSQL, SQL Server, SQLite and MongoDB, I design relational schemas, data dictionaries, metadata structures and tagging systems and develop SQL for reliable checking, updating and retrieval. Responsibility for DPUK's Great Minds Feasibility Tool includes troubleshooting the database scripts and scheduled processes that feed a Power BI dashboard used to assess cohort and data availability. I have also built Python- and SQLite-based integrity-monitoring workflows that maintain transactional records and flag unexpected changes across secure research storage. This experience is directly relevant to accurate data entry, database maintenance, discrepancy resolution and the dependable record keeping required throughout a longitudinal study.

Remote-monitoring data present their own challenges, including variable sampling, incomplete observations, temporal alignment and differences between source formats. To explore these issues in practice, I independently developed an open-source Multimodal Biomedical Preprocessing Pipeline for human-activity, EEG and ECG datasets. The workflow harmonises schemas and labels, segments time series, aligns metadata and produces consistent outputs, validation reports and manifests. HYPEHD, a separate public Python package, provides reusable analysis and visualisation tools for longitudinal and demographic health data. These projects complement my professional experience and provide a strong basis for learning the ART-transition measures and analysis plan. Python and R are my principal analytical tools for data manipulation, descriptive analysis, visualisation, feature engineering, classification, clustering and model evaluation.

Study Delivery, Organisation and Teamwork

Reliable longitudinal research depends on many small actions being completed accurately and at the right time. My current workload spans dataset onboarding, validation, secure provisioning, database and dashboard maintenance, analytical development, researcher requests, documentation, conference outputs and intern supervision. Priorities are managed according to deadlines, dependencies, data sensitivity and research risk, with Jira, Confluence, meeting records and technical documentation used to maintain continuity and accountability. This combination provides strong evidence of time management and organisational skills across concurrent responsibilities.

These habits transfer naturally to participant schedules, monitoring completion, debrief arrangements, database updates, newsletters, meeting actions and liaison with clinicians or collaborating partners. I am accustomed to checking whether expected information has arrived, resolving discrepancies, following issues across teams and escalating risks before they become delays. Supervision of an intern working on omics metadata discovery has further developed my ability to define a manageable scope, set weekly goals and acceptance criteria, review work constructively and support independent problem-solving. This would be valuable when guiding students or other staff contributing to the study.

Collaboration at DPUK spans researchers, clinicians, epidemiologists, statisticians, data owners, data managers, governance specialists and infrastructure engineers. The work often requires translation between a scientific question, the practical limitations of the available data and the technical or governance steps needed to proceed. At GSK, supervisors were based in the United States and findings were presented to colleagues from different countries and disciplinary backgrounds. Across both settings, effective teamwork has meant communicating early, adapting the level of technical detail and ensuring that decisions and responsibilities remain clear.

Communication, PPIE and Participant Support

Several strands of my work provide concrete evidence of communication beyond technical discussion. A researcher-facing course that I developed on secure HPC use, workflow execution and output checking turned unfamiliar processes into structured, practical guidance for users with varied computational experience. My first-author article, *Choosing the Right Genomic Dataset*, was conceptualised, organised and written by me with input and guidance from Professor Kenneth Langlands and Dr Fatemeh Torabi. Published by *Real World Data Science* in 2026, it translates complex questions of dataset selection, governance and analysis readiness into accessible guidance for researchers.

Public-facing workshops on AI safety, privacy and responsible use of sensitive health data have also required careful listening and explanation without jargon. Questions and concerns from participants were treated as substantive contributions, particularly where views differed about acceptable risk or the use of personal data. That experience has reinforced an approach well suited to PPIE and co-design: lived experience should shape the language, procedures and interpretation of a study rather than being consulted only after decisions have been made.

The same principles would guide my contact with the young people and other ART-transition participants. Consistent communication, clear expectations and respect for participants' time are central both to retention and to the validity of the resulting data. ADHD may affect organisation, communication and sustained engagement differently for each person, so support must remain patient, non-judgemental and responsive rather than assuming one approach will suit everyone. Accurate records, agreed contact and safeguarding procedures, appropriate boundaries and prompt escalation of concerns would underpin all participant interactions.

Building on this experience, I am ready to undertake the study-specific clinical responsibilities—including structured debrief interviews, support during remote monitoring and specimen processing—through formal training, supervision and approved protocols. Experience working to detailed SOPs, handling sensitive participant-derived information and documenting actions precisely provides a sound foundation for doing so safely. I would also complete the required DBS, occupational-health and research-passport processes promptly.

Contribution to ART-transition

I would bring Professor Kuntsi, Dr Bowler and the wider team an ability to contribute immediately to longitudinal-data cleaning and analysis, database management, study coordination, documentation and research outputs. My record includes a peer-reviewed clinical paper, a first-author methodological article, conference work on dementia subtyping and genomics infrastructure, a second-author privacy-aware NLP poster and contributions to DARE UK reports. These outputs reflect an ability to analyse carefully, write clearly, respond constructively to feedback and communicate limitations honestly.

Alongside these technical and research strengths, I would bring warmth, reliability and respect for the people who make the research possible. ART-transition's integration of participant relationships, day-to-day study delivery and longitudinal analysis is precisely what makes the post attractive to me. I would be proud to help participants remain informed and supported throughout the study, ensure that their data are handled with care and contribute to research that improves understanding of ADHD during a consequential stage of life.